US AIR QAULITY INDEX

The Air Quality Index (AQI) is a system used to communicate how polluted the air currently is or how polluted it is forecast to become. It helps the public understand the health effects associated with different levels of air pollution. The AQI focuses on key pollutants regulated by the Clean Air Act, including:

1. **Ground-level ozone (O3)**
2. **Particulate matter (PM10 and PM2.5)**
3. **Carbon monoxide (CO)**
4. **Sulfur dioxide (SO2)**
5. **Nitrogen dioxide (NO2)**

AQI Scale

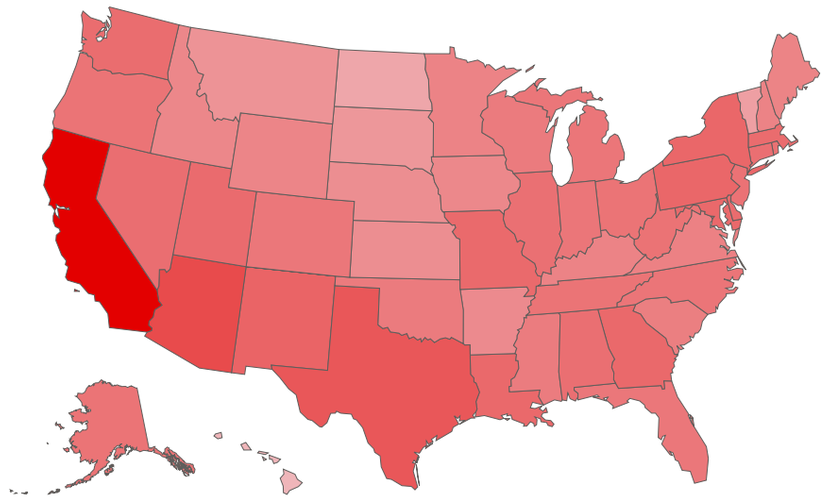
The AQI is divided into categories, each corresponding to a different level of health concern:

- **0-50**: Good (Green) – Air quality is considered satisfactory.
- **51-100**: Moderate (Yellow) – Air quality is acceptable; however, some pollutants may be a concern for a small number of people.
- **101-150**: Unhealthy for sensitive groups (Orange) – Members of sensitive groups may experience health effects, but the general public is not likely to be affected.
- **151-200**: Unhealthy (Red) – Everyone may begin to experience health effects; members of sensitive groups may experience more serious effects.
- **201-300**: Very Unhealthy (Purple) – Health alert; everyone may experience more serious health effects.
- **301-500**: Hazardous (Maroon) – Health warnings of emergency conditions; the entire population is more likely to be affected.

AQI Basics for Ozone and Particle Pollution

Daily AQI Color	Levels of Concern	Values of Index	Description of Air Quality
Green	Good	0 to 50	Air quality is satisfactory, and air pollution poses little or no risk.
Yellow	Moderate	51 to 100	Air quality is acceptable. However, there may be a risk for some people, particularly those who are unusually sensitive to air pollution.
Orange	Unhealthy for Sensitive Groups	101 to 150	Members of sensitive groups may experience health effects. The general public is less likely to be affected.
Red	Unhealthy	151 to 200	Some members of the general public may experience health effects; members of sensitive groups may experience more serious health effects.
Purple	Very Unhealthy	201 to 300	Health alert: The risk of health effects is increased for everyone.
Maroon	Hazardous	301 and higher	Health warning of emergency conditions: everyone is more likely to be affected.

Average of AQI by State Name



Reigons

All

US AIR QAULITY INDEX

Year

1980 2022

Average Sites Reporting

72

Average Population Density

2.07K

Records Reported

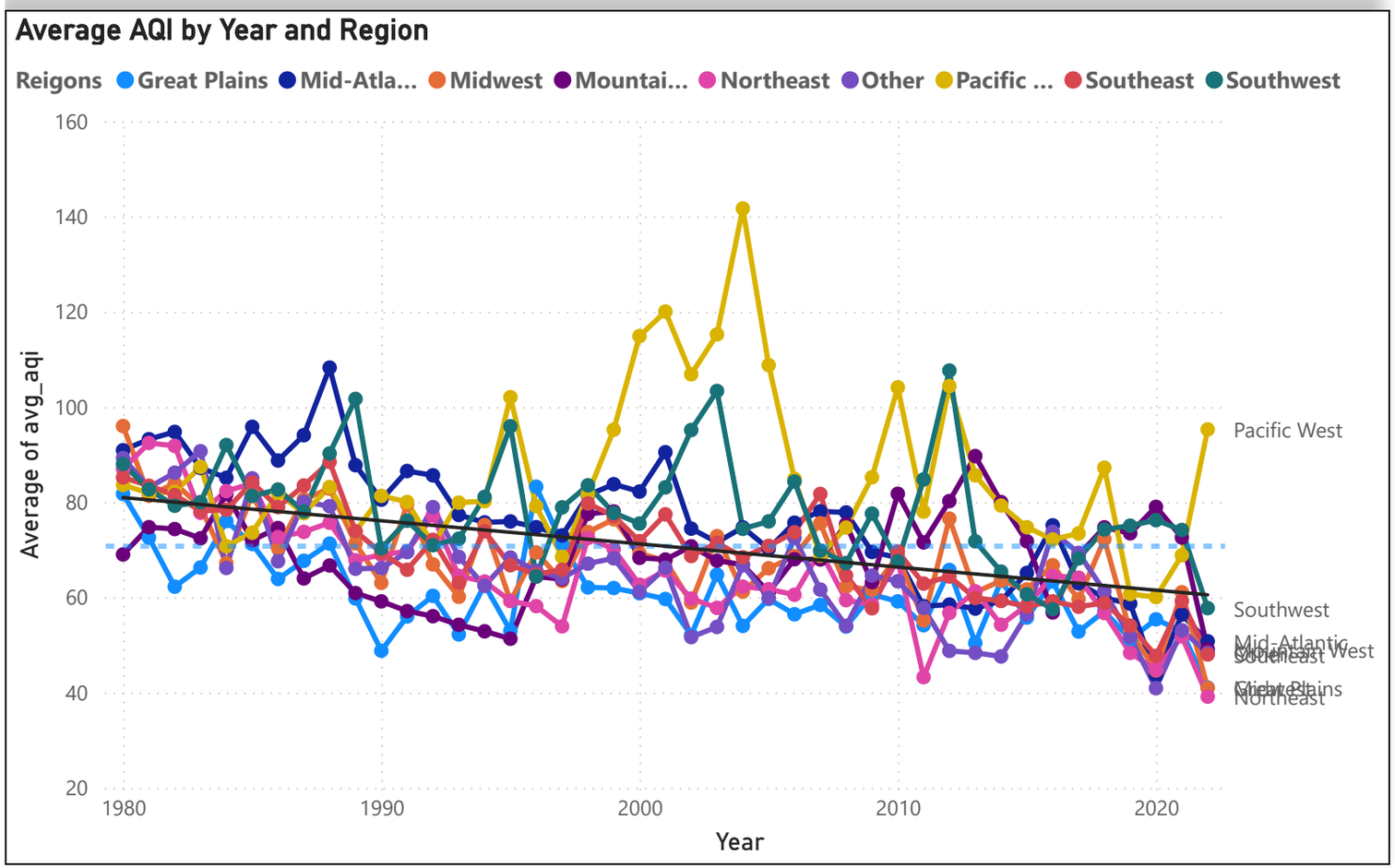
18.552K

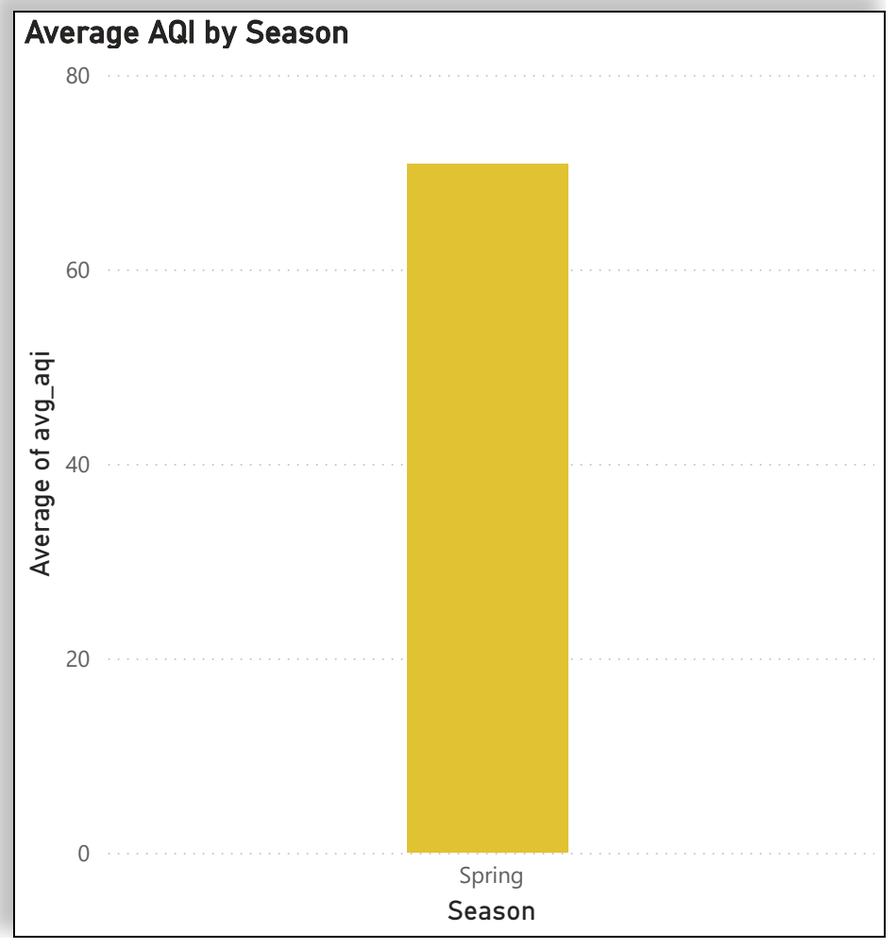
California

Worst_AQI_State

Season

Spring





Reigons

All

US AIR QAULITY INDEX

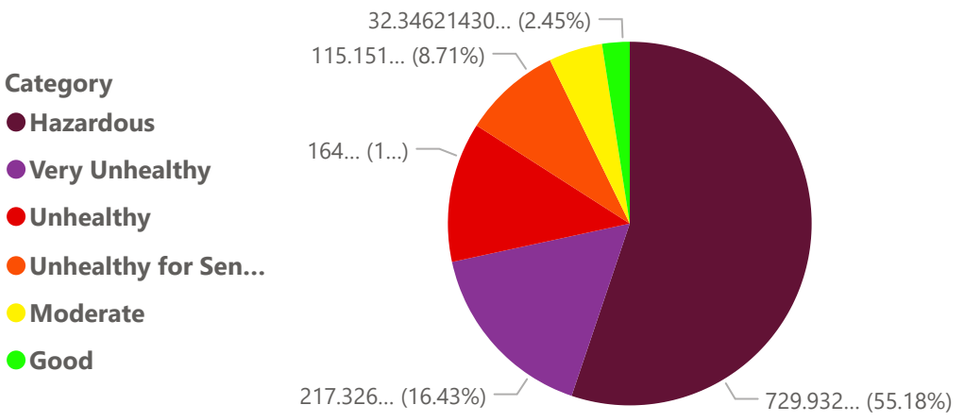
Year

1980 2022

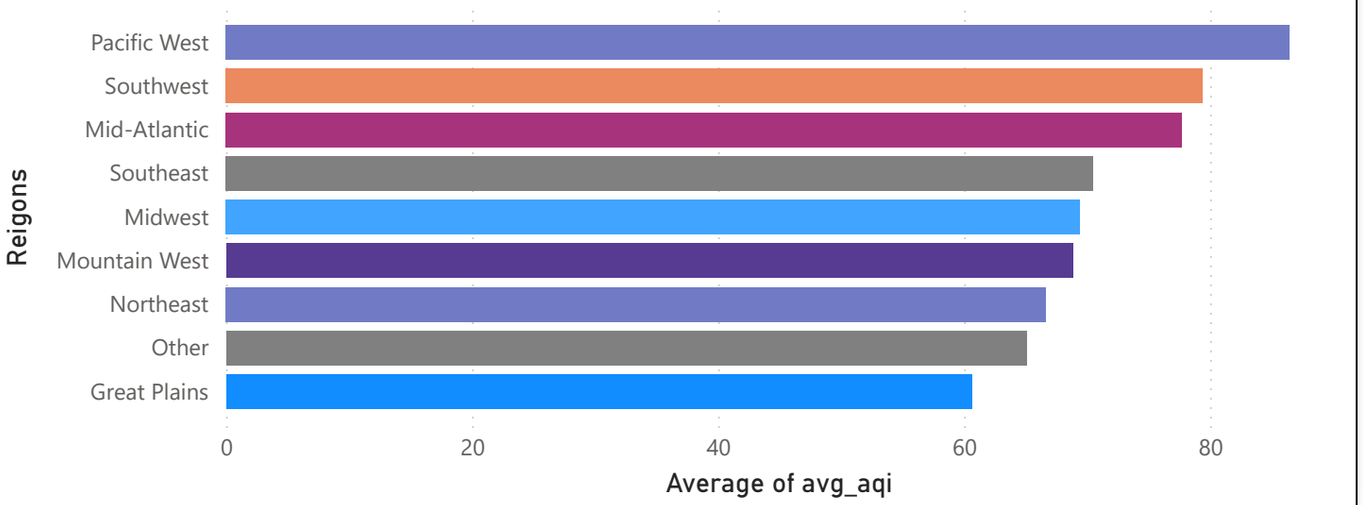
Season

Spring

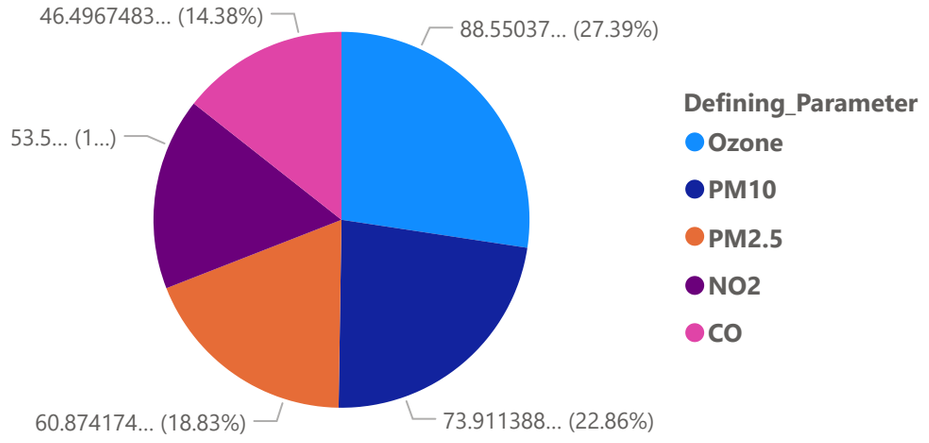
Average AQI by Category



Average AQI by Reigons



Average of AQI by Defining_Parameter



Top 5 Worst State according to AQI

state_name	Average of avg_aqi
California	135.86
Texas	88.24
Arizona	87.14
New Mexico	84.16
Pennsylvania	79.17
Total	97.73

Top 5 best State according to AQI

state_name	Average of avg_aqi
Hawaii	38.85
Montana	45.60
North Dakota	46.41
Puerto Rico	48.38
South Dakota	50.13
Total	46.40